

Simple Kenwood based requirements template — route is: User story and purpose → Use case and marketing requirements → technical requirements → specs (partialy execution)

1. Introduction and purpose of item or product

Device is intended to determine and log mainly the power consumption of small kitchen appliances while in real day use. The data will be analyzed and postprocessed to give statements about the power consumption (min, max, average, distribution in histograms) to improve the devices and to evaluate if the lifetime testing represents real use.

Implementation status summary (firmware v7 — Shelly + OTA + Power-loss)

|                                    |    |
|------------------------------------|----|
| ✓ Fulfilled                        | 21 |
| ⚠ Partial                          | 8  |
| ✗ Not fulfilled                    | 1  |
| 🔧 Hardware / out of firmware scope | 7  |
| i N/A (section header)             | 1  |

2. Use case / User story / Marketing requirement

| Req. ID | Detail                                                                                                                               | user requirement / tec req / spec | Source | Priority             | Verification method                        | Standard / test method | Status      | Notes / Evidence (firmware v7 — Shelly + OTA + Power-loss)                                                                                                                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------|----------------------|--------------------------------------------|------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U 1     | Device logs the power consumption of an attached slave device with common EU-plugs                                                   | user requirement                  | User   | shall / should / can | Visual inspection / Demonstration / Test / |                        | ✓ Fulfilled | U1: 230 V switching delegated to certified Shelly Plug S MTR Gen3 (EU F-type, CE-marked). ESP32 receives push from shelly_push.js via POST /api/shelly_push. ShellyClient:ingest() parses & validates.                                                                  |
| U 2     | It stores the data permanently                                                                                                       | user requirement                  | User   | shall                |                                            |                        | ✓ Fulfilled | U2: Samples stored on removable SD card as /log.csv in FILE_APPEND mode; no overwrite on boot (Logger.h).                                                                                                                                                               |
| U 3     | The stored data is postprocessed                                                                                                     | user requirement                  | User   | shall                |                                            |                        | ⚠ Partial   | U3: CSV format suitable for pandas/Excel. No dedicated post-processing scripts shipped yet — web UI provides liveplot and histogram for in-browser analysis.                                                                                                            |
| U 4     | The end user does not need to program or to set up the logger and just plugs it in                                                   | user requirement                  | User   | shall                |                                            |                        | ⚠ Partial   | U4: Plug-and-play after one-time factory pairing of the Shelly to S21 "PZEM_Logger". Once paired, the user only plugs both devices in. Degraded from ✓ to ⚠ vs. v4 due to the Shelly pairing step; mitigated by pre-pairing at manufacture.                             |
| U 5     | The use period ranges from 5 min to 50 h                                                                                             | user requirement                  | User   | shall                |                                            |                        | ⚠ Partial   | U5: Continuous operation supported (millis()) rollover ~49.7 days). Long-run (>50 h) not yet field-verified. Added risk vs. v4: the Wi-Fi link ESP32→Shelly must remain stable throughout.                                                                              |
| U 6     | It logs with appropriate sampling frequency and accuracy for standard small electric appliances such as hand mixers and juq blenders | user requirement                  | User   | shall                |                                            |                        | ⚠ Partial   | U6: Default 1 s push cadence from Shelly (was 0.5 s in v4 — Shelly meter updates at ~1 Hz). Adequate for kitchen appliances. Active power accuracy per Shelly datasheet. PF is derived (aspower / V * I), not hardware-measured — column renamed "pf" apparent" in CSV. |
| U 7     | It can be safely used in the kitchen environment                                                                                     | user requirement                  | User   | shall                |                                            |                        | 🔧 Hardware  | U7: Depends on enclosure build (IP44, cable glands, fuse, clearance/creepage). Not a firmware item — see specs section.                                                                                                                                                 |
| U 8     | It does not cost (materials only) more than 60 €                                                                                     | user requirement                  | User   | should               |                                            |                        | 🔧 Hardware  | U8: BOM-dependent. ESP32 + Shelly Plug S MTR Gen3 + SD shield + PSU + housing = 50–55 € in single-unit pricing; near but within the 60 € target.                                                                                                                        |

3. General requirements

|     |                                                                                                                        |         |           |        |                                |  |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|------------------------------------------------------------------------------------------------------------------------|---------|-----------|--------|--------------------------------|--|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Device polls power per default every 1 s                                                                               | tec req | U 6       | shall  |                                |  | ✓ Fulfilled     | Req 1.0: INTERVAL_SHELLY_POLL_MS = 1000 ms (Config.h v7). Shelly pushes at this cadence; Logger::poll(Due()) consumes the cached value. Changed from 0.5 s (v4) — Shelly internal meter is 1 Hz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2   | Device logs power with affiliated timestep per default every 10 s to memory                                            | tec req | U 6       | shall  |                                |  | ✓ Fulfilled     | Req 2.0: INTERVAL_SD_FLUSH_MS = 10000 ms in Config.h. Logger::flush(Due()) writes RAM buffer to SD every 10 s. Each sample carries its own millis() timestamp.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3   | Device does not overwrite existing log data                                                                            | tec req | U 2       | shall  |                                |  | ✓ Fulfilled     | Req 3.0: Log file is opened with FILE_APPEND (Logger::flush(rSSD). Reset only via explicit POST /reset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4   | The memory is removable / interchangeable                                                                              | tec req |           | shall  |                                |  | 🔧 Hardware      | Req 4.0: Standard SD card on SPI — physically removable (PIN_SDCS=25 etc.).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5   | Device uses a format easy to post process on PC - for instance CSV or similar                                          | tec req | U 3       | shall  |                                |  | ✓ Fulfilled     | Req 5.0: CSV header "time,ms,voltage,V,power,W,pf,apparent" (LOG_FILE_HEADER in Config.h v7). Column renamed from "pf" to "pf_apparent" to reflect its derived nature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6   | memory resists 100 h which equals 36 000 k write cycles at 1/10s writing frequency                                     | tec req | U 5 / U 2 | shall  |                                |  | ✓ Fulfilled     | Req 6.0: SD cards withstand >100 k write cycles per block; 36 000 appends to one file is well within spec. Flush every 10 s (not 1/10 s), so the actual write load is 10× lower than the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7   | It is capable to measure up to 10 A at 230 V                                                                           | tec req | U 6       | shall  |                                |  | 🔧 Hardware      | Req 7.0: Shelly Plug S MTR Gen3 rated 16 A / 3680 W max (per datasheet); covers the 'up to 10 A' requirement. Effective range limited by external 8–10 A fuse per BOM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 9   | Device transmits logged data via wireless function to host                                                             | tec req | U 4       | should |                                |  | ✓ Fulfilled     | Req 9.0: WiFi module (WiFi_AP_STA) — ESP32 runs softAP for phone/laptop; Shelly joins same AP as STA client. POST /api/shelly_push receives measurements every 1 s. No external network                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 10  | Allows to connect both types of electric devices: protective earth devices and EU-plug devices                         | tec req | U 1       | shall  |                                |  | 🔧 Hardware      | Req 10.0: Depends on socket/plug wiring (grounded EU socket per BOM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11  | Fulfills the requirements for clearance and creepage electric protection is >= IP44                                    | tec req | U 7       | shall  |                                |  | 🔧 Hardware      | Req 11.0: Enclosure / PCB layout responsibility — not firmware.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 12  | allows safe flush of memory if the power connection is determined                                                      | tec req | U 7       | should |                                |  | ⚠ Partial       | Req 12.0: Enclosure responsibility (IP44 housing + cable gland per BOM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 13  | allows safe flush of memory if the power connection is determined                                                      | tec req | U 7       | should |                                |  | ⚠ Partial       | Req 13.0: Mains-loss detection implemented in PowerMonitor.h (firmware v7). GPIO 35 reads the 9 V rail via a 180k/47k divider; on a sustained drop below 7.35 V the sketch flushes the RAM buffer to SD (Logger::flushToSD), sheds load (Wi-Fi + LED off) and idles until brownout or mains recovery. 7.35 V chosen so the worst-case ADC error stays above the TSR-1-2450 -6.5 V regulation cliff. ADC robustness: analogReadMilliVolts (w/ use calibration) + 16× oversampling + 3-of-N majority voting + 0.4 V hysteresis; 3 s startup grace for supercap (TSR-1-2450). ⚠ Partial: pending bench validation of timing/threshold — estimated ~95 % flush success in nominal conditions, lower for rapid power- |
| 14  | User interface: The device comes with a user interface that                                                            | tec req | N/A       |        |                                |  | ⚠ N/A           | Section header (UI requirement group).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 15  | - blinks 1/s if polling and logging work correctly and no wireless connection is established                           | tec req |           | shall  |                                |  | ✓ Fulfilled     | Req 15.0: INTERVAL_LED_OK_MS = 500 ms toggles the LED at 1 Hz (StatusLed: update with LED_OK).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 16  | - blinks 5/s in case polling or logging do not work correctly                                                          | tec req |           | shall  |                                |  | ✓ Fulfilled     | Req 16.0: INTERVAL_LED_ERR_MS = 100 ms → LED_ERROR at 5 Hz. Triggered by SD error OR Shelly watchdog timeout (SHELLY_ERROR, THRESHOLD = 3 × 1000 ms = 3 s silence). Logger::ok() now checks shellyOk() && sdOk(). NOTE: LED will also blink 5 Hz during OTA upload — expected, recovers automatically on reboot.                                                                                                                                                                                                                                                                                                                                                                                                 |
| 16b | - blinks 0.5/s if polling and logging work correctly and wireless connection is established                            | tec req |           | can    |                                |  | ✗ Not fulfilled | Req: 0.5 Hz blink when a wireless client is connected is NOT implemented. StatusLed only has two states (LED_OK / LED_ERROR). ACTION: add a third state LED_OK_WIFI and switch on WiFi: softAPpos(SStationNum)) > 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 17  | Device comes with an additional on-line user interface once connected to host via wireless that                        | tec req |           | should |                                |  | ✓ Fulfilled     | Req 17.0: Web portal served from softAP at default IP 192.168.4.1 (WebPortal: begin).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 18  | - displays the current live data on the host                                                                           | tec req |           | should |                                |  | ✓ Fulfilled     | Req 18.0: Home page polls /api/live every 1 s; shows power (W), voltage (V), pf_apparent, JSON also includes shelly ok, sd ok, and ota active fields. Label updated "pf" → "pf_apparent".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 19  | - allows to download all stored data - in the "DOWNLOAD DATA" section                                                  | tec req |           | should |                                |  | ✓ Fulfilled     | Req 19.0: GET /download streams /log.csv as Content-Disposition: attachment (WebPortal: handleDownload).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 20  | - allows to delete / reset the data storage memory - in the "DELETE MEMORY" section                                    | tec req |           | should |                                |  | ✓ Fulfilled     | Req 20.0: POST /reset calls Logger::resetSDFile() (deletes log.csv and re-creates with header). Confirmation dialog in JS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 21  | - allows to set the poll rate between one poll every 1, 2, 5, 10, 30 s - in the "SETUP" section                        | tec req |           | can    |                                |  | ⚠ Partial       | Req 21.0: /settings page rate grid allows {1000, 2000, 5000, 10000, 30000} ms. handleApiSettingsSave() whitelist enforces this. Sub-second options (200/500 ms) removed — Shelly meter is 1 Hz. Status ⚠ because rates other than 1/s are not yet validated end-to-end against the fixed 1 Hz Shelly push cadence.                                                                                                                                                                                                                                                                                                                                                                                               |
| 22  | - allows to set up the start of log at power consumptions greater than 1, 2, 5, 10, 20, 50 W                           | tec req |           | can    |                                |  | ✓ Fulfilled     | Req 22.0: Threshold grid (0, 1, 2, 5, 10, 20, 50) W in /settings UI. handleApiSettingsSave() whitelists value; Logger::setPowerThreshold() applies immediately. Threshold lost on reboot (RAM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 23  | - allows the user to read a built in user manual - "MANUAL"                                                            | tec req |           | can    |                                |  | ⚠ Partial       | Req 23.0: /readme page (PAGE_README) updated to reflect Shelly architecture and includes OTA section. Still light on troubleshooting depth — basic operations covered, no advanced setup                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 24  | - allows to plot the live data on the host - "LIVE PLOTTER"                                                            | tec req |           | can    |                                |  | ✓ Fulfilled     | Req 24.0: /liveplot canvas oscilloscope shows last 60 samples. fetchIntervalMs synced to /api/settings, floored at 1000 ms (Shelly meter cadence). Auto-scales Y axis to peak.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|     |                                                                                                                        |         |           |        |                                |  |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 27  | device should carry all relevant information for wireless connectivti (ID, password,...) on the device                 | tec req |           | should |                                |  | ✓ Fulfilled     | Req 27.0: Housing labelled with WiFi name (PZEM_Logger), password (logger1234), web UI URL (http://192.168.4.1 or http://braun_pzem.local), and Shelly URL (http://192.168.4.2).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 28  | Device must not connect to the internet but locally only                                                               | tec req |           | should |                                |  | ✓ Fulfilled     | Req 28.0: WiFi mode (WiFi_AP_STA) — softAP only as anchor for Shelly and user. No WAN route. Shelly cloud disabled at factory. STA mode reserved for future NTP if needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 28b | It is accessed by a domain that is plain text - BRAUN_PZEM - and not by hard to remember urls like: http://192.168.4.1 | tec req |           | can    |                                |  | ✓ Fulfilled     | Req 28a.0: MDNS.begin(WiFi_AP_HOSTNAME) in WebPortal: begin(). Reachable at http://braun_pzem.local on iOS, macOS, and most Android. Previously open in v4 — code already existed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 29  | Firmware updates can be installed over the air (OTA) without a physical USB connection to the ESP32                    | tec req | U 4       | can    | Demonstration / browser upload |  | ✓ Fulfilled     | Req 29.0: OTA implemented via ESP32 Arduino Update.h. GET /update serves an upload form (PAGE_OTA) with live progress bar; POST /update streams the .bin to Update.write() chunk-by-chunk and reboots on success. Logger::setOtaInProgress(true) flushes the RAM buffer to SD and pauses polling before flashing — no log samples lost. Logging resumes automatically after reboot. SD-card retention of log.csv is unaffected (different memory).                                                                                                                                                                                                                                                               |

4. Technical specs

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |  |     |  |  |             |                                                                                                                                                                                                                                                                                                                                                                                   |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--|-----|--|--|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Use existing off-the-shelf components where possible:<br>- Shelly Plug S MTR Gen3 (replaces PZEM-004T + CT + EU socket + AC/DC print PSU)<br>- ESP32<br>- certified 230 V → 5 V PSU (+ TSR-1-2450 9 V→5 V on the rail)<br>- SD card reader shield<br>- grounded EU-plug<br>- empty off-the-shelf plastic housing<br>- cable gland<br>- LED + resistor<br>- switch + resistor<br>- wago spring terminals/connectors<br>- overcurrent fuse 6–10 A<br>- AC power-loss detection (2×1 F supercaps + | spec |  | can |  |  | ⚠ Partial   | BOM: Shelly Plug S MTR Gen3 + certified 230–5 V PSU (TSR-1-2450 on the 9 V rail) + ESP32 + SD shield + IP44 housing + 2×1 F supercaps + 180k/47k divider for power-loss sensing. Removed vs. v4: PZEM-004T, CT, EU socket, AC/DC print PSU. Capacitor-based power-loss circuit now in place and handled in firmware (Req 13). LED and pushbutton still pending in hardware build. |
|  | PIN layout ESP32 (Config.h v7):<br>GPIO 16 / 17 — frei (ehem. PZEM UART2, entfällt mit Shelly)<br>GPIO 25 — SD CS<br>GPIO 14 — SD MOSI<br>GPIO 27 — SD CLK<br>GPIO 26 — SD MISO<br>GPIO 32 — LED<br>GPIO 33 — Taster (reserviert, noch nicht im Code)<br>GPIO 35 — ADC Versorgungüberwachung (9 V-Rail, Power-loss, Req 13)                                                                                                                                                                     | spec |  | can |  |  | ✓ Fulfilled | Pin layout per Config.h v7: SD CS/MOSI/CLK/MISO = 25/14/27/26, LED = 32. GPIO 35 ADC now actively read by PowerMonitor for 9 V-rail power-loss detection (Req 13) via 180k/47k divider. GPIO 33 reserved for button (future, intentionally unconfigured). GPIO 16/17 (former PZEM UART2) freed and unassigned. No 230 V connections anywhere on the ESP32 side.                   |